CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 8

NMAP LIVE HOST DISCOVERY

AIM

To locate active hosts on a network using Nmap discovery methods, including ARP, ICMP, TCP, and UDP scans.

OBJECTIVE

To learn how Nmap detects responding systems before carrying out more detailed network scans.

SOFTWARE REQUIRED

• Nmap

• TryHackMe Nmap room / AttackBox

• A computer with network connectivity

INTRODUCTION

Nmap, short for Network Mapper, is a free, open-source utility for scanning networks, discovering active hosts, and identifying network services. This practical covers four host-discovery methods: ARP, ICMP, TCP, and UDP.

The purpose of host discovery is to find which systems are responding before performing a more detailed scan. Nmap documentation describes ARP as useful for targets on a local Ethernet network, while ICMP, TCP, and UDP probes provide other ways to check whether a host is reachable.

1. ARP HOST DISCOVERY

ARP sends a request to find the MAC address associated with a target IP address on the same local Ethernet network. Receiving a reply suggests that the target is reachable on that local network.

```
Command: nmap -PR -sn TARGETS
```

ARP discovery is generally limited to the local subnet because ARP operates at the link layer.

Figure 1. Basic ARP host-discovery concept.

2. ICMP HOST DISCOVERY

Nmap may send an ICMP Echo Request (Type 8) and check for an ICMP Echo Reply (Type 0). A reply indicates that the target is responding to the probe.

```
Command: nmap -PE -sn TARGETS
```

Other ICMP discovery options mentioned in the lab are -PP for ICMP Timestamp and -PM for ICMP Address Mask requests.

Figure 2. Basic ICMP Echo discovery concept.

3. TCP SYN HOST DISCOVERY

TCP SYN Ping sends a TCP probe with the SYN flag set to a chosen port. Responses such as SYN/ACK or RST can indicate that the target host is responding.

```
Command: nmap -PS23 TARGETS
```

The lab material notes that TCP SYN Ping does not require a privileged account in the described context, whereas TCP ACK Ping does.

Figure 3. Basic TCP SYN discovery concept.

4. UDP HOST DISCOVERY

UDP Ping sends a UDP probe to a selected port. If the port is closed, the target may return an ICMP Port Unreachable message, which can be evidence that the host is active.

```
Command: nmap -PU40125 TARGETS
```

Figure 4. Basic UDP discovery concept.

IMPORTANT NMAP COMMANDS

Purpose	Option
Discover hosts only	nmap -sn TARGETS
ARP discovery	nmap -PR -sn TARGETS
ICMP Echo	nmap -PE -sn TARGETS
ICMP Timestamp	nmap -PP -sn TARGETS
ICMP Address Mask	nmap -PM -sn TARGETS
TCP SYN Ping	nmap -PS23 TARGETS
TCP ACK Ping	nmap -PA80 TARGETS
UDP Ping	nmap -PU40125 TARGETS
Turn off reverse DNS	nmap -n TARGETS
Perform reverse DNS lookup	nmap -R TARGETS

OBSERVATION

1. ARP can identify systems on the same local subnet.
2. ICMP Echo can be used to test whether a host replies.
3. TCP SYN Ping checks a host by sending a SYN probe to a TCP port.
4. TCP ACK Ping sends an ACK probe; in the lab context, it requires a privileged account.
5. UDP Ping may reveal an active host through an ICMP Port Unreachable reply from a closed UDP port.
6. The -sn option performs host discovery without continuing into a port scan.

OBSERVATION TABLE

Scan Type	Option	Expected observation
ARP	-PR	ARP reply / MAC information from a local host
ICMP Echo	-PE	ICMP Echo Reply
ICMP Timestamp	-PP	ICMP Timestamp Reply
ICMP Address Mask	-PM	ICMP Address Mask Reply
TCP SYN	-PS	SYN/ACK or RST response
TCP ACK	-PA	RST response may indicate a live host
UDP	-PU	ICMP Port Unreachable may indicate a live host

RESULT

Thus, Nmap live-host discovery was studied using ARP, ICMP, TCP, and UDP techniques.

CONCLUSION

Nmap supports several ways to find active hosts. ARP is mainly suitable for systems on the same local Ethernet network, while ICMP, TCP, and UDP probes offer additional methods to check whether a target responds.

DIAGRAM NOTE

The four figure captions represent simplified educational concepts based on the protocol behavior described in the supplied practical. They are not screenshots from another website.

Reference: Nmap Network Scanning – Host Discovery documentation.